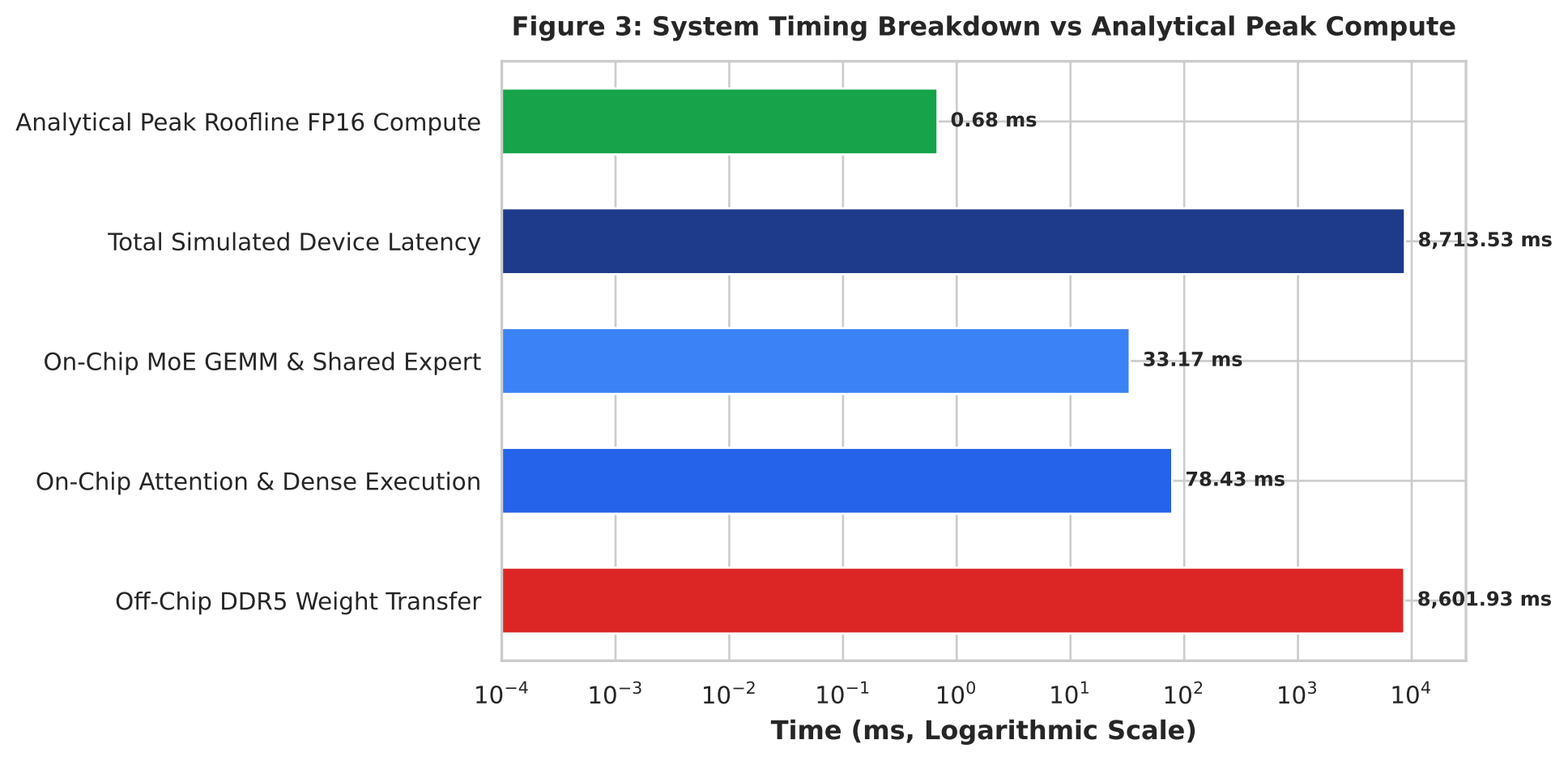
Figure 3: System Timing Breakdown vs Analytical Peak Compute
Analytical Peak Roofline FP16 Compute
0.68 ms
Total Simulated Device Latency
8,713.53 ms
On-Chip MoE GEMM & Shared Expert
33.17 ms
On-Chip Attention & Dense Execution
78.43 ms
Off-Chip DDR5 Weight Transfer
8,601.93 ms
10^{-4}
10^{-3}
10^{-2}
10^{-1}
10^{0}
10^{1}
10^{2}
10^{3}
10^{4}
Time (ms, Logarithmic Scale)